

Tian Lan

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Education

- 2013–2017 PhD in Theoretical Physics
Perimeter Institute for Theoretical Physics & University of Waterloo
Thesis: [A Classification of \(2+1\)D Topological Phases with Symmetries](#)
Supervisor: Xiao-Gang Wen
- 2012–2013 MSc in Theoretical Physics, Perimeter Scholars International (PSI)
Perimeter Institute for Theoretical Physics & University of Waterloo
- 2008–2012 BSc in Mathematics and Physics, Tsinghua University

Experience

- 2018–2020 Postdoctoral Fellow
Institute for Quantum Computing, University of Waterloo
Supervisors: Raymond Laflamme, Bei Zeng
- 2021.6–now Assistant Professor
The Chinese University of Hong Kong

Research Interests

My research focuses on the theory of topological phases of matter: intrinsic topological order, symmetry protected topological phases (including topological insulators as a special case), symmetry enriched topological phases, higher dimensional generalization of symmetry and topological phases, and their mathematical foundation: tensor category, higher category, higher algebra, group cohomology, algebraic topology, etc. I obtained the first complete classification of 3+1D topological orders and the techniques used can also greatly simplify the study of topological phases in even higher dimensions. Recently, topological defects have been recognized as generalized symmetries, which is a conceptual revolution. I am interested in the underlying mechanism and mathematical framework for generalized symmetry, in order to better understand and organize phases and phase transitions.

Grants

PI, Early Career Scheme (ECS) No. 24304722, Hong Kong Research Grants Council
PI, Guangdong Provincial Quantum Science Strategic Initiative No. GDZX2501013
Co-PI, Collaborative Research Fund (CRF) No. C7015-24GF, Hong Kong Research Grants Council

Teaching

2021-25 Fall PHYS5410 Advanced Quantum Mechanics
2024-26 Spring PHYS1111 University Physics I

Referee

Physical Review Series (PRX, PRX Quantum, PRL, PRResearch, PRB)
Communications in Mathematical Physics, Journal of High Energy Physics, SciPost Physics, Quantum ...

Associate Editor

Advances in Theoretical and Mathematical Physics.

Talks

- A “Periodic Table” of Topological Orders

- SIAM Annual Meeting (AN16), Boston, Massachusetts, USA, July 2016 (invited but was not able to attend due to visa delay)
- Department of Physics, Fudan University, May 2018
- Department of Physics, Tsinghua University, July 2018
- Graduate School of China Academy of Engineering Physics, July 2018
- Department of Physics, Southern University of Science and Technology, July 2019
- The Chinese University of Hong Kong, October 2020

- Non-Abelian Family of Topological Order²¹

- Croucher Summer Course “Quantum Entanglement and Topological Order”, The Chinese University of Hong Kong, June 2017
- Department of Physics, Southern University of Science and Technology, July 2019

- Matrix Formulation of Non-Abelian Family¹³

- Topological Quantum Computing (TQC2019), Shenzhen, China, December 2019

- Higher Dimensional Topological Order, Higher Category and A Classification in $3+1D$ ^{14,17}

- Department of Physics, Tsinghua University, May 2018
- Séminaire de mathématiques supérieures (SMS) 2018, Fields Institute for Research in Mathematical Sciences, June 2018
- Quantum Information and Operator Algebras III, Academy of Mathematics and Systems Science, Chinese Academy of Sciences, July 2018
- Department of Physics, Sun Yat-Sen University, July 2019
- QuIST V: The 5th International Conference on Quantum Information, Space-time, and Topological order, Kunming, China, August 2019
- Department of Physics, Southern University of Science and Technology, August 2019
- The Chinese University of Hong Kong, October 2020
- CMSA, Harvard University, June 2021
- BIMS, March 2022

- The Modular Extension Characterization of SPT/SET Phases^{19,20,23}

– BIMSA, December 2021

- Introduction to Categorical Approach to Topological Phases in Arbitrary Dimensions

– Center of Mathematical Sciences and Applications, Harvard University, April 2020

- Higher symmetry enriched topological phases^{10,11}

– Topological Orders and Higher Structures, Erwin Schrödinger Institute for Mathematics and Physics, University of Vienna, August 2020

- Quantum Current⁶

– Croucher Summer Course "Quantum Entanglement and Topological Order", CUHK, June 2023

– The first International Congress of Basic Science (ICBS), BIMSA, July 2023

– Entangled Quantum Matter Workshop, Tsinghua University, August 2023

- Gauging of generalized symmetry/Category of SET orders⁵

– Mini-Workshop on Unconventional Superconductivity and Correlated Electron Systems, CUHK, Dec 2023

– The International Congress of Basic Science (ICBS), BIMSA, July 2024

– Applications of Generalized Symmetries and Topological Defects to Quantum Matter, Simons Center for Geometry and Physics, September 2024

- Tube category, tensor renormalization and topological holography (Origin of topological holography)²

– Tensor Network States: Algorithms and Applications, CUHK, December 2024

– The University of Hong Kong, June 2025

– SUSTech Summer School on Generalized Symmetry and Quantum Criticality, October 2025

– MIST Conference, CUHK, February 2026

- Pro-Tensor Network¹

– Advanced Numerical Methods for Highly Entangled Quantum Matter, CY Cergy Paris Universite, France, May 2026

– Frontier Workshop on Topological States and Strongly Correlated Quantum Matter, Tsung-Dao Lee Center of Sciences and Arts, Dali, July 2026

Publications

- [1] Gen Yue, Ansi Bai, Linqian Wu, and **Tian Lan**, "Pro-tensor network", (2026), [arXiv:2605.06661](#).
- [2] **Tian Lan**, "Tube category, tensor renormalization and topological holography", [Communications in Mathematical Physics](#) **406**, 196 (2025), [arXiv:2412.07198](#).
- [3] Gen Yue, Longye Wang, and **Tian Lan**, "Condensation completion and defects in 2+1d topological orders", [Physical Review B](#) **112**, 104106 (2025), [arXiv:2402.19253](#).
- [4] Ying Chan, **Tian Lan**, and Linqian Wu, "Torus algebra and logical operators at low energy", (2024), [arXiv:2403.01577](#).
- [5] **Tian Lan**, Gen Yue, and Longye Wang, "Category of SET orders", [Journal of High Energy Physics](#) **2024**, 111 (2024), [arXiv:2312.15958](#).
- [6] **Tian Lan** and Jing-Ren Zhou, "Quantum current and holographic categorical symmetry", [SciPost Physics](#) **16**, 053 (2024), [arXiv:2305.12917](#).
- [7] Chenqi Meng, Xinping Yang, **Tian Lan**, and Zhengcheng Gu, "Non-invertible spts: an on-site realization of (1+1)d anomaly-free fusion category symmetry", (2024), [arXiv:2412.20546](#).
- [8] Wenjie Xi, **Tian Lan**, Longye Wang, Chenjie Wang, and Wei-Qiang Chen, "On a class of fusion 2-category symmetry: condensation completion of braided fusion category", [Advances in Theoretical and Mathematical Physics](#) **29**, 151–204 (2023), [arXiv:2312.15947](#).
- [9] Wenjie Xi, Ya-Lei Lu, **Tian Lan**, and Wei-Qiang Chen, "A lattice realization of general three-dimensional topological order", (2021), [arXiv:2110.06079](#).
- [10] Liang Kong, **Tian Lan**, Xiao-Gang Wen, Zhi-Hao Zhang, and Hao Zheng, "Algebraic higher symmetry and categorical symmetry: a holographic and entanglement view of symmetry", [Physical Review Research](#) **2**, 043086 (2020), [arXiv:2005.14178](#).

- [11] Liang Kong, **Tian Lan**, Xiao-Gang Wen, Zhi-Hao Zhang, and Hao Zheng, “Classification of topological phases with finite internal symmetries in all dimensions”, *Journal of High Energy Physics* **2020**, 93 (2020), [arXiv:2003.08898](#).
- [12] **Tian Lan**, Xueda Wen, Liang Kong, and Xiao-Gang Wen, “Gapped domain walls between 2+1D topologically ordered states”, *Physical Review Research* **2**, 023331 (2020), [arXiv:1911.08470](#).
- [13] **Tian Lan**, “Matrix formulation for non-Abelian families”, *Physical Review B* **100**, 241102(R) (2019), [arXiv:1908.02599](#).
- [14] **Tian Lan** and Xiao-Gang Wen, “Classification of 3+1D bosonic topological orders (II): the case when some pointlike excitations are fermions”, *Physical Review X* **9**, 021005 (2019), [arXiv:1801.08530](#).
- [15] **Tian Lan**, Chenchang Zhu, and Xiao-Gang Wen, “Fermion decoration construction of symmetry-protected trivial order for fermion systems with any symmetry and in any dimension”, *Physical Review B* **100**, 235141 (2019), [arXiv:1809.01112](#).
- [16] Chenchang Zhu, **Tian Lan**, and Xiao-Gang Wen, “Topological nonlinear σ -model, higher gauge theory, and a systematic construction of 3+1D topological orders for boson systems”, *Physical Review B* **100**, 045105 (2019), [arXiv:1808.09394](#).
- [17] **Tian Lan**, Liang Kong, and Xiao-Gang Wen, “Classification of 3+1D bosonic topological orders: the case when pointlike excitations are all bosons”, *Physical Review X* **8**, 021074 (2018), [arXiv:1704.04221](#).
- [18] **Tian Lan**, “A classification of (2+1)D topological phases with symmetries”, PhD thesis (University of Waterloo, Sept. 2017), p. 86, [arXiv:1801.01210](#).
- [19] **Tian Lan**, Liang Kong, and Xiao-Gang Wen, “Classification of (2+1)-dimensional topological order and symmetry-protected topological order for bosonic and fermionic systems with on-site symmetries”, *Physical Review B* **95**, 235140 (2017), [arXiv:1602.05946](#).
- [20] **Tian Lan**, Liang Kong, and Xiao-Gang Wen, “Modular extensions of unitary braided fusion categories and 2+1D topological/SPT orders with symmetries”, *Communications in Mathematical Physics* **351**, 709–739 (2017), [arXiv:1602.05936](#).
- [21] **Tian Lan** and Xiao-Gang Wen, “Hierarchy construction and non-Abelian families of generic topological orders”, *Physical Review Letters* **119**, 040403 (2017), [arXiv:1701.07820](#).
- [22] Keren Li, Yidun Wan, Ling-Yan Hung, **Tian Lan**, Guilu Long, Dawei Lu, Bei Zeng, and Raymond Laflamme, “Experimental identification of non-Abelian topological orders on a quantum simulator”, *Physical Review Letters* **118**, 080502 (2017), [arXiv:1608.06932](#).
- [23] **Tian Lan**, Liang Kong, and Xiao-Gang Wen, “Theory of (2+1)-dimensional fermionic topological orders and fermionic/bosonic topological orders with symmetries”, *Physical Review B* **94**, 155113 (2016), [arXiv:1507.04673](#).
- [24] **Tian Lan**, Juven C. Wang, and Xiao-Gang Wen, “Gapped domain walls, gapped boundaries, and topological degeneracy”, *Physical Review Letters* **114**, 076402 (2015), [arXiv:1408.6514](#).
- [25] **Tian Lan** and Xiao-Gang Wen, “Topological quasiparticles and the holographic bulk-edge relation in (2+1)-dimensional string-net models”, *Physical Review B* **90**, 115119 (2014), [arXiv:1311.1784](#).